

MANAGING FOREIGN EXCHANGE RISKS

As developments in communications and transportation have increased accessibility to other parts of the world, companies have become increasingly involved in conducting business in foreign markets. Because each country has its own medium of exchange (currency), companies involved in international business must deal with a new dimension not present in domestic activities—that of managing foreign exchange risks. These risks occur because the exchange rates between currencies fluctuate, causing a change in the value of the company's international activities in terms of its domestic currency.

There are three principal ways that foreign exchange fluctuations affect a company's financial performance:

1. By changing the expected results of transactions denominated in non-domestic currencies (i.e., “transaction” exposure).
2. By changing the domestic currency value of net assets held in foreign currencies (i.e., “translation” exposure).
3. By changing the domestic currency value of future cash flows to be earned in foreign currencies, or by changing a company's future competitive position (i.e., “economic” exposure).

Each of these exposures results from different corporate activities and is determined in a different way.

Through astute management (and perhaps a little luck), it is possible to minimize the foreign exchange risks associated with conducting business in a global environment. This note discusses the nature of the three foreign exchange risks and some methods companies might use to manage them.

Assessing Foreign Exchange Risks

Transaction exposure

Transaction exposure exists during the normal course of international business transactions whenever two or more currencies are involved and there is a lag between the date a contract is signed or goods delivered and the date of payment. Transaction exposure also exists when there is a time difference between the declaration of a dividend by a foreign subsidiary and its actual payment. The risk associated with transaction exposure is that the exchange rate will fluctuate before the transaction is completed and the currency exchange occurs, thereby changing the amount of domestic currency paid or received. It is important to note that transaction exposure arises from specific transactions.

Transaction exposure would exist, for example, if a U.S. electronics company contracted to sell 10,000 semiconductors to a French computer manufacturer at a sales price of F10 (French Franc) per unit with delivery in 90 days and payment due on delivery. If the exchange rate of US\$ to F on the date of the contract signing were US\$.23/F, then the U.S. company would expect to receive US\$23,000 for the sale (i.e., $F10 \times 10,000 \times \text{US}\$.23/\text{F} = \text{US}\$23,000$). If the exchange rate of US\$ to F on the date of delivery were US\$.21/F, then the U.S. company would receive only US\$21,000.

Translation exposure

Translation exposure occurs because of the need to translate the financial statements of foreign subsidiaries and affiliates into the currency of the parent company in order for consolidated financial statements to be prepared. As no exchange of cash or other assets takes place, translation exposure is solely the result of the financial accounting and reporting process.

U.S. companies are required to translate foreign financial statements in accordance with the standards set forth in Statement of Financial Accounting Standards No. 52 (FAS 52), *Foreign Currency Translation*, issued by the Financial Accounting Standards Board (FASB) in December 1981. FAS 52 superseded the highly criticized and controversial FAS 8 that had been in effect only about six years. According to FAS 52, the objectives of translation are to (1) provide information that is generally compatible with the expected economic effects of an exchange rate change on an enterprise's cash flows and equity, and (2) reflect in consolidated financial statements the financial results and relationships of the individual consolidated entities as measured in their *functional currencies* in conformity with U.S. generally accepted accounting principles. Additionally, the Board expressed the belief that, to the extent practicable, accounting for foreign currency translation in the United States should harmonize with related practices followed in other industrialized countries around the world.

As noted in the translation objectives, FAS 52 adopts a “functional currency” approach to translation. This is a radical departure from the “U.S. dollar” approach taken in FAS 8, because it permits multiple measurement bases in consolidated financial statements. The FAS 52 approach is

based on the premise that the foreign entities comprising a multinational enterprise operate and generate cash flows in a variety of economic environments and that differences in these environments should be recognized in the translation process. Thus, the first step is to identify the functional currency for each entity included in the financial statements of the reporting enterprise. An entity's *functional currency* is defined as the currency of the primary economic environment in which it operates; normally, that is the currency of the environment in which an entity primarily generates and expends cash. Although FAS 52 provides guidance regarding the determination of the functional currency, considerable judgment will often be necessary. As the resulting translated financial statements can differ significantly, selection of each foreign entity's functional currency is an important managerial decision.

Once the functional currencies have been determined, the accounts of each foreign entity should be brought into conformity with U.S. generally accepted accounting principles and *remeasured* in its functional currency, with any gains or losses generally treated as adjustments to income. (Remeasurement is not necessary if the foreign financial statements are already stated in the entity's functional currency.) If the functional currency is the U.S. dollar, the foreign entity's financial statements should be remeasured into U.S. dollars. With minor exceptions, the remeasurement process into a U.S. dollar functional currency is the same as the translation process previously required by FAS 8: the temporal method is used and any gains or losses resulting from the remeasurement process are reflected in income.

If the functional currency is other than the U.S. dollar, the remeasured financial statements are *translated* into U.S. dollars using the *current rate method*. Any translation gains or losses are taken directly to a separate component of stockholders' equity and have no effect on current income. Translation from the functional currency to the U.S. dollar is not necessary when the U.S. dollar is the functional currency because the remeasurement process previously described converts foreign financial statements into U.S. dollars.

Although the objectives and procedures of foreign currency translation are vastly different under FAS 52 from those of FAS 8, when the U.S. dollar is determined to be the functional currency, the translation process produces similar results to those that would have resulted under FAS 8. **Exhibit 1** contains a summary of some of the major objections to FAS 8 and the corresponding provisions contained in FAS 52.

Table 1 provides a comparison of the current rate and temporal methods of foreign currency translation. As can be seen, a company's translation exposure is quite different under these two methods. Under the temporal method (which is essentially a monetary-nonmonetary approach) exposure depends on the relationship between a company's monetary assets and its monetary liabilities. Under the current rate method, a company's exposure depends on the relationship between its total assets and total liabilities. As companies frequently have net monetary liability positions yet typically have net total asset positions, translation exposure under the current rate method is often the reverse of that under the temporal method.

Table 1

A Comparison of Balance Sheet Translation Methods

Balance Sheet Item	Current Rate Method	Temporal Method
Cash	Current	Current
Receivables	Current	Current
Inventory	Current	Historical ¹
Fixed assets	Current	Historical
Payables	Current	Current
Long-term debt	Current	Current
Owners' equity	Historical	Historical
¹ Unless the lower of cost or market (LCM) rule results in inventory being stated at market, then <i>current</i> .		
NOTE: <i>Current</i> means that the item is translated at the rate that existed at the balance sheet date. <i>Historical</i> means that the item is translated at the exchange rate that existed at the time the asset was acquired, the debt issued, or the capital contributed.		

As an example of balance sheet translation in accordance with FAS 52, assume that the same U.S. electronics company previously discussed established a French subsidiary with the following balance sheet:

French Subsidiary
(000s)

<u>Assets</u>		<u>Liabilities and Equity</u>	
Cash	F 50	Current liabilities	F 350
Inventory (cost)	300	Long-term debt	50
Plant and Equipment (net)	<u>100</u>	Equity	<u>50</u>
Total	<u>F 450</u>	Total	<u>F 450</u>

With the initial exchange rate of US\$.23/F, the translated values of the subsidiary would be as follows.

French Subsidiary
(000s)

<u>Assets</u>		<u>Liabilities and Equity</u>	
Cash	US\$ 11.5	Current liabilities	US\$ 80.5
Inventory (cost)	69.0	Long-term debt	11.5
Plant and Equipment (net)	<u>23.0</u>	Equity	<u>11.5</u>
Total	<u>US\$103.5</u>	Total	<u>US\$103.5</u>

If the French franc subsequently weakened against the dollar with the exchange rate falling to US\$.21/F and the balance sheet of the French subsidiary remained exactly the same in French francs, the translated values would then be:

French Subsidiary
(000s)

<u>Assets</u>		<u>Liabilities and Equity</u>	
Cash	US\$ 10.5	Current liabilities	US\$ 73.5
Inventory (cost)	63.0	Long-term debt	10.5
Plant and Equipment (net)	<u>21.0</u>	Equity	<u>10.5</u>
Total	<u>US\$ 94.5</u>	Total	<u>US\$ 94.5</u>

Although the values attached to the subsidiary's assets and liabilities remain constant when stated in French francs, the translation process produces a loss of US\$1,000, the amount of the decrease in the equity account needed to balance the subsidiary's U.S. dollar balance sheet. Assuming the French franc to be the functional currency, the translation loss would appear as a separate component of equity in the consolidated balance sheet.

This example illustrates that a U.S. company's translation exposure is a function of the relationship between a foreign entity's assets and liabilities, because all such accounts are translated at current exchange rates. A loss occurs whenever the foreign currency weakens against the U.S.

dollar. Conversely, a gain occurs whenever the foreign currency strengthens against the U.S. dollar. Thus, FAS 52 attempts to capture the economic consequences of the change in exchange rates.

The amount of the translation adjustment can also be calculated directly by multiplying a company's exposed asset or liability position by the amount of the currency exchange rate fluctuation. In the preceding example, the French subsidiary had a net asset position of FF 50 determined as follows:

Total Assets	F 450
Less: Total Liabilities	<u>400</u>
Position (long)	F 50

The loss would be calculated as $F 50 \times US\$(.02)/F = (US\$1,000)$.

In addition to changing the manner in which foreign balance sheets are translated, FAS 52 also changed the way that foreign income statements are translated. When a foreign entity's local currency is the functional currency, application of the current rate method required by FAS 52 results in translating both cost of sales and depreciation at the weighted average exchange rate for the period. Under the temporal method required by FAS 8, these two important expenses were translated at historical rates. In the case of the French subsidiary just described, application of FAS 52 would result in smaller translated amounts for cost of sales and depreciation than would have been the case under FAS 8.

In terms of financial reporting disclosures, FAS 52 requires the following:

- The aggregate *transaction* gain or loss included in determining net income for the period. This includes those resulting from the remeasurement process and those resulting from forward contracts.
- An analysis of the current year's changes in the separate component of equity reflecting cumulative translation adjustments.
- If significant, disclosure of the effects of a rate change subsequent to the date of the financial statements on unsettled balances pertaining to foreign currency transactions may be necessary.

Economic exposure

Economic exposure can be thought of as comprising two separate but potentially related components: "future exposure" and "competitive exposure." *Future exposure* deals with the impact on the domestic currency value of an income stream denominated in a different currency when exchange rates change. For example, Figure 1 shows the yen(¥)/U.S. dollar(\$) exchange rate for the period 1978-89. The yen strengthened considerably during this period, and the exchange rate ranged from a low of about 250 yen per U.S. dollar in 1984 to a high of about 123 yen per U.S. dollar in

1987. That is, it took ¥250 to buy \$1 in 1984, while in 1987 it took only ¥123 to buy \$1. An income stream of ¥250,000,000 each year for 20 years would have had a dollar equivalent of about \$1,000,000 with an exchange rate of ¥250/\$ but \$2,032,520 with an exchange rate of ¥123/\$. Thus, the value in terms of the domestic currency of the cash flow can be dramatically affected by changes in the exchange rate. Future exposure depends on how much of a change in domestic currency value results from a change in the exchange rates.

The measure of future exposure is the difference between the present value of the cash flows determined at the current rates and the present value at the expected future rates. Evaluation of this exposure is useful in determining what might happen if the exchange rates moved, or after the fact in evaluating the changes in the subsidiary's economic value following a change in the currency value. (See **Exhibit 2.**)

The other form of economic exposure, *competitive exposure*, affects a potentially broader segment of business. Many companies do not have a foreign currency denominated source of cash flow, but they do have foreign competitors. The impact on the ability of a domestic company to compete in the domestic economy can be drastically affected by changes in exchange rates when a major competitor is a foreign company. For example, Caterpillar Tractor's main competitor during the 1980s was Kamatsu of Japan, and its operating results were dramatically affected by fluctuations in the exchange rate.

Figure 2 shows both the ¥/\$ exchange rates and Caterpillar's sales during the period 1978-89. It illustrates the competitive advantage that Kamatsu enjoyed when the yen was weak relative to the dollar. With an exchange rate of ¥250/\$1, Kamatsu could sell its products in the U.S. relatively cheaply in dollar terms and still receive an acceptable return in terms of yen. As the yen strengthened in the mid-1980s, Caterpillar's sales recovered. In order to maintain the same yen income, Kamatsu would have had to double its U.S. dollar prices as the exchange rate fell from ¥250/\$ in 1984 to ¥123/\$ by the end of 1987. Thus, the competitive position of a company can be dramatically affected by changes in exchange rates. This loss or increase in competitiveness can affect the value of the company. It is this phenomenon that is referred to as competitive exposure.

The function of the foreign exchange markets

All three types of foreign exchange risks (i.e., transaction, translation, and economic) result from movements in the exchange rates or parity among currencies. Because these changes occur in the currency markets, an understanding of the role and function of the foreign exchange markets is critical to the successful management of foreign exchange risks. Foreign exchange markets exist only for convertible currencies. For these currencies, the exchange rates generally are determined by market forces without substantial governmental interventions. The currencies of all established industrialized and some newly industrialized countries are convertible.

Figure 1

MANAGING FOREIGN EXCHANGE RISKS

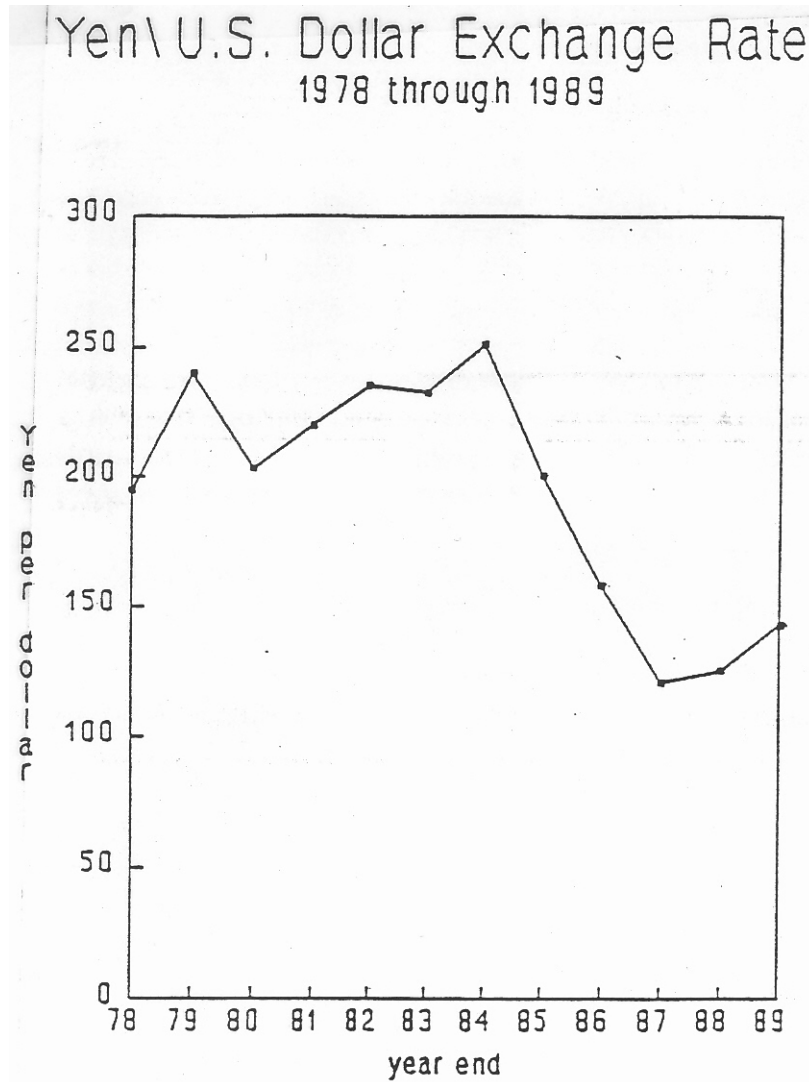
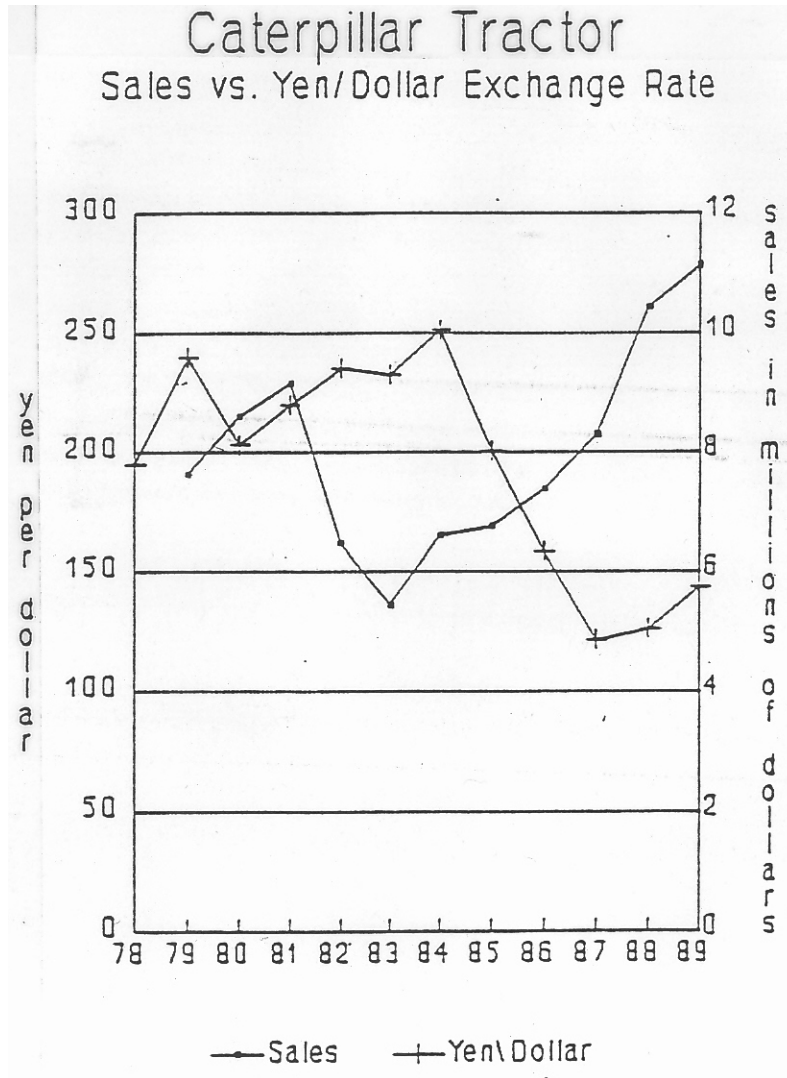


Figure 2

MANAGING FOREIGN EXCHANGE RISKS



Trading in the foreign exchange markets proceeds through contracts made between buyers and sellers of foreign currency. Three types of exchange markets exist, each distinguished by the type of contract involved. These are the *spot*, *forward*, and *futures* markets. In a spot market, the parties agree to exchange one currency for another at a fixed ratio to be delivered immediately, but more likely in one to two business days. Typically, individuals and companies exchange at the spot rate through banks. The spot exchange rate is set based on trading among banks for immediate exchange (i.e., allowing one or two days for clearing).

In the forward market, the buyer and seller agree to exchange currency at a fixed ratio at some specified future date. Nothing happens until that date, at which time the exchange is made at the agreed upon exchange rate. In general, forward contracts are between a bank and a buyer and/or a seller. Each of these contracts is designed by the bank to meet the specific needs of the other party with regard to amounts and timing. Traditionally, forward contracts are negotiated for delivery in one, two, three, six, and twelve months, although for major currencies (e.g., U.S. dollar, Japanese yen, British pound, West German mark, Swiss franc, French franc, and Canadian dollar) future delivery dates of up to three years are possible.

Futures markets are organized markets for trading contracts for the future delivery of currency. They differ from the forward markets in that the contracts are standardized in terms of time of delivery and size of contract. For example, each futures contract on the West German mark (Deutsche Mark) traded on the International Monetary Market (IMM) in Chicago has a standardized size of DM125,000. The contracts available for trading have different expiration dates depending on the time of year. For example, on May 15, 1990 the available contract expiration dates on the West German mark were June 1990, September 1990, December 1990, and March 1991. An additional feature of futures contracts is that they may require the exchange of cash before the delivery date. The disadvantage of futures contracts over forward contracts is that the standardized features may not exactly match a company's requirements. Forward contracts can be designed to match those requirements. On the other hand, the standardization of futures contracts can make them less costly.

Table 2 reports quotes for spot, forward, and futures markets for British pounds and West German marks. The first quote under Exchange Rates is the spot rate of 1.6740 dollars per pound. The subsequent quotes are forward rates for 30, 90, and 180 days from the current date. Futures prices are reported in the second half of the table. The first line for each currency specifies the size of the contract and below that are listed the contracts that are available for trading and the most recent exchange rates at which the contracts were traded.

The spot rates are the prevailing market rates determined by the supply and demand for a particular currency relative to another currency. Forward and futures rates differ from spot rates in that they are rates for exchange at future dates and usually are quoted at a premium or discount to spot rates. The size of a currency's premium or discount depends on the financial communities' assessment of the future of the relative strength of that currency. This assessment is based on such factors as the country's balance of payments, trade figures, international reserves, GNP, money supply, rate of inflation, difference in interest rates, and other domestic and international economic indicators.

Table 2

MANAGING FOREIGN EXCHANGE RISKS**Currency Trading***Wall Street Journal*, May 16, 1990**Exchange Rates**

Tuesday, May 15, 1990

The New York foreign exchange selling rates below apply to trading among banks in amounts of \$1 million and more as quoted at 3 p.m. Eastern time by Bankers Trust Co. Retail transactions provide fewer units of foreign currency per dollar.

Country	U.S. \$ equiv.		Currency per U.S. \$	
	<u>Tues.</u>	<u>Mon.</u>	<u>Tues.</u>	<u>Mon.</u>
Britain (pound)	1.6740	1.6815	.5974	.5983
30-day forward	1.6643	1.6715	.6004	.5983
90-day forward	1.6465	1.6539	.6073	.6046
180-day forward	1.6204	1.6281	.6171	.6142
West German (mark)	.6086	.6089	1.6430	1.6422
30-day forward	.6088	.6091	1.6427	1.6418
90-day forward	.6088	.6091	1.6427	1.6418
180-day forward	.6086	.6091	1.6431	1.6417

Futures

	<u>Open</u>	<u>High</u>	<u>Low</u>	<u>Settle</u>	<u>Change</u>
British Pound (IMM) ¹ - 62,500 pds.; \$ per pound					
June	1.6768	1.6784	1.6614	1.6630	-.0082
Sept	1.6506	1.6514	1.6340	1.6362	-.0080
W. GermanMark (IMM) - 125,000 marks; \$ per mark					
June	.6082	.6090	.6061	.6085	-.0008
Sept	.6082	.6088	.6062	.6084	-.0008
Dec	.6072	.6080	.6065	.6081	-.0009
Mar-91	.6073	.6073	.6071	.6078	-.0012

¹IMM - International Monetary Market at the Chicago Mercantile Exchange.

Quite often, especially outside the United States, forward and futures exchange rates are expressed in terms of an annualized percentage deviation from the spot rate. This is the case in the *Financial Times*. The following equation is used:

$$\frac{FR - SR}{SR} \times \frac{12}{N} \times 100\% = \text{Discount (or premium) in percentage terms}$$

where FR = Forward or future rate (local currency per unit of foreign currency)
 SR = Spot rate (local currency per unit of foreign currency)
 N = Months to maturity.

For example, a U.S. purchaser of a 90-day forward contract in British pounds would be buying at a 6.57 percent discount.

$$\frac{1.6465 - 1.6740}{1.6740} \times \frac{12}{3} \times 100\% = -6.57\%$$

On the other hand, a 90-day forward contract for the West German mark would be purchased with a .131 percent premium,

$$\frac{.6088 - .6086}{.6086} \times \frac{12}{3} \times 100\% = .131\%$$

The spot market serves immediate currency needs, whereas the forward and futures markets provide the means of dealing with foreign exchange risks.

Managing Foreign Exchange Risks

Managing transaction risks

The first two approaches described below are examples of ways of creating riskless hedges. Choosing between using a forward contract and a money market hedge is simple. Select the one that is the least expensive. The third and fourth approaches (leading and lagging, and invoicing currency) serve to reduce but not to eliminate the risk. The fifth approach (swaps) can be an effective way of eliminating risk but may be difficult to implement.

Hedging Through Foreign Exchange Market Contracts: A hedge in the foreign exchange market simply requires the agreement of two parties to exchange currencies at a stated rate (known when the contract is made) at some future point in time. This forward exchange rate will differ from the present spot rate as previously explained. The contract will cover an exposed transaction by guaranteeing the exchange rate at which the remittance will be made. (See **Exhibit 3**.) Hedging can

also be done with futures contracts but it is more difficult to match the amount and timing of the transaction.

Hedging Through Money Market Loans: A money market hedge is essentially an immediate conversion of loaned funds. Funds are borrowed by the payor in local currency and immediately converted to the recipient's currency (and usually placed in an interest-bearing account). In this manner, the effect of future changes in the exchange rates is avoided altogether. When payment is due, the remittance is transferred from the interest-bearing account to the recipient. Meanwhile, the payor needs only to pay off the original funds loaned (plus interest). (See **Exhibit 4**.)

Leading and Lagging: Leading and lagging the timing of cash exchanges is one of the simplest ways to control foreign exchange exposure. This involves speeding up, even prepayment, and slowing down the exchange of funds between companies in order to transfer the funds at relatively advantageous (to the exposed party) exchange rates given forecasts of future rates. It should be noted that many countries have imposed restrictions on the amount of time that funds can be led or lagged. Such laws seriously impair a manager's ability to apply this technique successfully.

Invoicing Currency: In some situations, it may be advantageous for a company to bill customers in a currency other than the customer's local currency to alleviate transaction exposure. Such a technique is especially useful to companies that have a major customer in one country and a major supplier in another. In this case, the customer might be billed in the supplier's currency.

To a lesser extent this technique may be an aid to decreasing translation exposure as well as enabling the company to hold currencies other than its own local currency, particularly in the case where one is an appreciating currency and the other is a depreciating currency.

Swaps: Loan swaps, currency swaps, and credit swaps are all basically the same; they allow companies to provide financing to a foreign affiliate without having to channel the funds through the foreign exchange market. A swap entails dual, offsetting loans of equal value (at the start of the arrangement) in two different currencies between two companies or institutions. Normally, each consolidated company involved will be a loan recipient in one currency and the lender in the other. (See **Exhibit 5**.)

Currency swaps are similar to the loan swaps described in **Exhibit 5** except that no loans are recorded. They are viewed simply as an exchange of one currency for another. In a credit swap, one of the companies involved will be a bank with a foreign office ready to make a loan to a company's subsidiary upon deposit of equivalent currency value in the bank's domestic location.

Managing translation risks

Balance Sheet Adjustments: This technique involves maintaining a net zero exposure (or if desired, a net asset or liability exposure) through the management of account balances. Obviously, such a strategy is limited by the necessity of integrating this technique with normal operations;

ideally, astute management will lower translation exposure without interfering with the normal transactions.

Licensing: In some cases, where a foreign affiliate may retain high exposure to the detriment of the parent company, it may be wise for the company to consider licensing its product to an independent foreign company rather than maintaining a subsidiary. Licensing eliminates all translation exposure because there is no affiliate to consolidate and, depending upon the licensing agreement, may alleviate all transaction exposure as well. Ideally, licensing should have been carefully considered prior to the establishment of the foreign affiliate.

Managing economic risks

These types of risks are the most difficult to manage because the exposure is long term and the risks are not associated with known events or timing. The best that can be done is to minimize these risks without substantially or unknowingly increasing the risks in other areas. For example, a company that produces a product domestically and sells it in a foreign market has its sales revenue exposed to economic risk. By moving production to the foreign market and incurring production costs in the foreign currency, only the profit margin would have economic exposure. Such a move may, however, create new transaction and/or translation exposure. These are the kinds of tradeoffs that must be considered in managing economic risks.

Conclusion

The size of the company, which includes its diversity of holdings and international activities, dictates the magnitude of its foreign exchange risks. The previous examples of managing risk continually emphasized the necessity of integrating various techniques with the normal operations of the company. All of the techniques have costs attached to them. Some costs are measurable while others are often difficult to measure. Only when the costs of the techniques are less than the impact of the undesired exposure will the techniques be worthwhile. Once the decision is made to manage foreign exchange risks, the approaches used will be affected by the size and diversity of foreign operations. Coordinated corporate action is generally preferable to independent action by individual subsidiaries and affiliates.

Exhibit 1

MANAGING FOREIGN EXCHANGE RISKS

**A Summary of the Major Objections to FAS 8 and the
Corresponding Provisions of FAS 52**

Major Objections to FAS 8

1. Use of the temporal method resulted in a translation (financial reporting) exposure that was inconsistent with the concurrent economic exposure.
2. Caused misleading and undesirable fluctuations in income as a result of requiring the inclusion of translation (paper) gains and losses, over which management had no control, in determining current period income.
3. Was too restrictive in its recognition of economic hedges of foreign exchange risk.
4. Distorted normal relationships within the financial statements of foreign divisions or subsidiaries.

Corresponding Provisions of FAS 52

1. Companies are required to determine the *functional currency* for each of their foreign entities. The functional currency is the currency of the primary economic environment in which the entity operates and generates net cash flows. For an entity with operations that are relatively self-contained and integrated within a particular foreign country, the functional currency generally would be the currency of that country. For an entity with operations not primarily self-contained, however, the functional currency might not be the currency of the country in which the entity is located. If a foreign entity's operations are a direct and integral component or extension of its U.S. parent company, then the U.S. dollar would most likely be the foreign entity's functional currency.

Unlike FAS 8 which was based on the premise that all foreign financial statements of a U.S. enterprise should be measured in U.S. dollars and treated remeasurement and translation as a simultaneous process, FAS 52 makes an important distinction between them. *Remeasurement* is the process of measuring in the functional currency those accounts that are denominated in another currency. *Translation* is the process of converting financial statements expressed in the functional currency into U.S. dollars. If the functional currency is determined to be the U.S. dollar, then remeasurement is accomplished in a manner very similar to the translation process required by FAS 8. If the functional currency is a foreign currency, then remeasurement is necessary whenever the accounts are stated in a foreign currency other than

Exhibit 1 (continued)

the functional currency. (Remeasurement is not necessary if the accounts are already stated in the functional currency.) With certain exceptions, adjustments resulting from the remeasurement process are made to income. Once remeasurement is accomplished, the foreign financial statements are translated into U.S. dollars using the *current rate method*. (Translation is not necessary if the functional currency is the U.S. dollar because the foreign financial statements are converted into U.S. dollars as a result of the remeasurement process.) Special provisions are made in FAS 52 for foreign operations in highly-inflationary economies. Whenever the cumulative three-year inflation rate is approximately 100 percent or more, the U.S. dollar should be used as if it were the functional currency.

2. Translation gains and losses resulting from the translation of a foreign entity's financial statements expressed in its functional currency into those expressed in U.S. dollars (using the current rate method) are no longer included in current period income but are to be accumulated in a separate component of consolidated equity until sale or until complete or substantially complete liquidation of the net investment in the foreign entity occurs.

3. Criteria for recognizing forward exchange contracts as foreign currency hedges are more flexible than those of FAS 8. In addition, FAS 52 acknowledges that foreign currency transactions other than forward exchange contracts can serve as hedges of foreign currency commitments. The relaxation of the criteria applicable to hedge accounting allows companies to defer many of the gains and losses previously included in income. At the time the related commitment transactions are recognized, the amount of the exchange gains or losses previously deferred are treated as adjustments to the measurement bases of the transactions.

4. The concept of functional currency and the use of the current rate method permit more normal relationships within the translated financial statements of foreign entities.

Exhibit 2

MANAGING FOREIGN EXCHANGE RISKS

Economic Exposure

The general formulation for calculating the economic exposure is as follows:

$$\text{Economic Exposure} = PV_o - PV_a$$

$$PV_o = \frac{CF_{o1} \times R_{o1}}{1 + K} + \frac{CF_{o2} \times R_{o2}}{(1 + K)^2} \dots + \frac{CF_{on} \times R_{on}}{(1 + K)^n} \quad 4$$

$$PV_a = \frac{CF_{a1} \times R_{a1}}{1 + K} + \frac{CF_{a2} \times R_{a2}}{(1 + K)^2} \dots + \frac{CF_{an} \times R_{an}}{(1 + K)^n} \quad 5$$

Where:

PV_o =Present value of the cash flows at the original exchange rates

PV_a =Present value of the cash flows at the adjusted or new exchange rates

CF_{on} =Cash flow from the subsidiary in year n with the original exchange rates

CF_{an} =Cash flow from the subsidiary in year n with the adjusted or new exchange rates

R_{on} =Original exchange rate in year n

R_{an} =Adjusted or new exchange rate in year n

K =Discount Rate

Exhibit 2 (continued)

Using the example of the U.S. electronics company, assume that at the time of the investment, the U.S. company expected its French subsidiary to generate the following cash flows.

Year 1	F100,000
Year 2	120,000
Year 3	130,000
Year 4 and continuing through Year n	150,000

Assuming a discount rate of 15 percent for the French subsidiary, at the exchange rate of US\$.23/F, the present value of the cash flows to the U.S. parent company would be US\$191,890. This would be the economic value of the French subsidiary.

With the strengthening of the dollar against the French franc, the expected cash flows might be reduced. This could occur because the French subsidiary would have to pay more in terms of French francs for the raw materials imported to manufacture the semiconductors. Because of an increase in raw material costs, the expected cash flows might be reduced to the following:

Year 1	F 95,000
Year 2	115,000
Year 3	120,000
Year 4 and continuing through Year n	140,000

With the exchange rate changed to US\$.21/F, the present value of the cash flows using the same 15 percent discount rate would be US\$164,120. The economic value of the subsidiary was reduced by US\$27,770 by the revaluation of the U.S. dollar.

Exhibit 3

MANAGING FOREIGN EXCHANGE RISKS

Hedging Through Foreign Exchange Market Contracts

Using the example of the U.S. electronics company, assume the following sequence of events:

May 15, 1990: Deliver F100,000 worth of semiconductors to a French computer manufacturer, (F=French franc; sales price = F10 per unit). Payment is due on August 13, 1990 in French francs. On May 15, 1990, spot prices of F5.545/US\$ (US\$0.18034/F), the export sale is worth US\$18,034 if no credit terms are extended. In order to protect against an adverse change in the value of the F against the US\$, the U.S. company sells F100,000 against the US\$ for delivery on August 13, 1990, at F5.5625/US\$ (US\$0.17978/F), the market price for three-month dollars against the French franc. The forward contract will provide \$17,978.

August 13, 1990: The U.S. company receives a check for F100,000. The U.S. company delivers F100,000 against the forward contract and receives US\$17,978.

From the beginning of the transaction, the U.S. company knows the exact amount of dollars it will receive when the French franc payment is finally realized. By using the forwards contract the company is guaranteed to exchange Fs at the rate of F5.5625 per US\$ or US\$0.17978 per F, thus eliminating the exchange risk.

Exhibit 4

MANAGING FOREIGN EXCHANGE RISKS

Hedging Through Loans

Again, using the example of the U.S. electronics company, assume the following sequence of events:

May 15, 1990: The U.S. company delivers the F100,000 worth of semiconductors to the French computer manufacturer. Payment is due on August 13, 1990, in French francs. On May 15, the spot price of US\$.18034 would result in an export sale worth US\$18,034. In order to protect against an adverse change of the Franc against the dollar, the U.S. company borrows F100,000 from a Paris bank @ 9 percent on a discount basis¹, converts the F97,800 proceeds from the loan into U.S. dollars, $(F97,800 \times US$.18034 = US\$17,637)$, and invests the dollars in a three-month Treasury Bill in the U.S. @ 8.00 percent. After three months the bill would be worth US\$17,990.

August 13, 1990: The U.S. company receives a check for FF100,000 from the French company. The French franc revenues from the export sale are used to pay the F100,000 loan from the Paris bank. The three-month T-bill is liquidated and it yields the anticipated US\$17,990. By undertaking this series of actions, the U.S. company knows the exact amount of dollars it will receive when the French franc payment is finally realized.

¹A discount basis is a loan which at maturity, three months in this case, requires a payment of principal and interest exactly equal to F100,000. In order for this to be true, the original loan amount must be F97,800, interest is $F97,800 \times \frac{.09}{4} = F2200$, for a total F100,000 due on August 13, 1990.

Exhibit 5

MANAGING FOREIGN EXCHANGE RISKS

Hedging Through Swaps

The structure of back-to-back, or parallel loans (i.e., swaps) might be as follows:

May 15, 1990

SPOT RATE US\$0.18034/F F5.545/US\$

The French subsidiary of a U.S. parent needs to borrow F100,000 with the intention to repay the F100,000 to the parent company on August 13, 1990. At the same time, the U.S. subsidiary of a French parent needs to borrow US\$18,034 with the intention of repaying the parent company on August 13, 1990. Individually, each parent company would incur transaction exposure, since its subsidiary will repay the loan in local currency. The U.S. parent will receive F100,000 and the French parent will receive \$18,034. However, by the U.S. parent agreeing to lend \$18,034 to the U.S. subsidiary of the French parent and the French parent agreeing to lend F100,000 to the French subsidiary of the U.S. parent, each company can eliminate its exchange rate risk. Both loans are due on August 13, 1990, and the result on that date would be:

U.S. parent — \$18,034 from the U.S. subsidiary of the French parent
French parent — F100,000 from the French subsidiary of the U.S. parent

If the loan agreement contained a rollover provision, that is, the loan would be extended for another three months, the exchange rate on August 13, 1990 would be used to adjust the loan principal. For example, if the exchange rate were \$0.19/F on August 13, 1990, the U.S. dollar would have weakened against the French franc and the French parent's loan to the French subsidiary of the U.S. parent would have a dollar equivalent of \$19,000. In this case, the U.S. parent would have to advance an additional \$966 to the U.S. subsidiary of the French parent in order to bring the principal balances to equivalent values as of August 13, 1990.